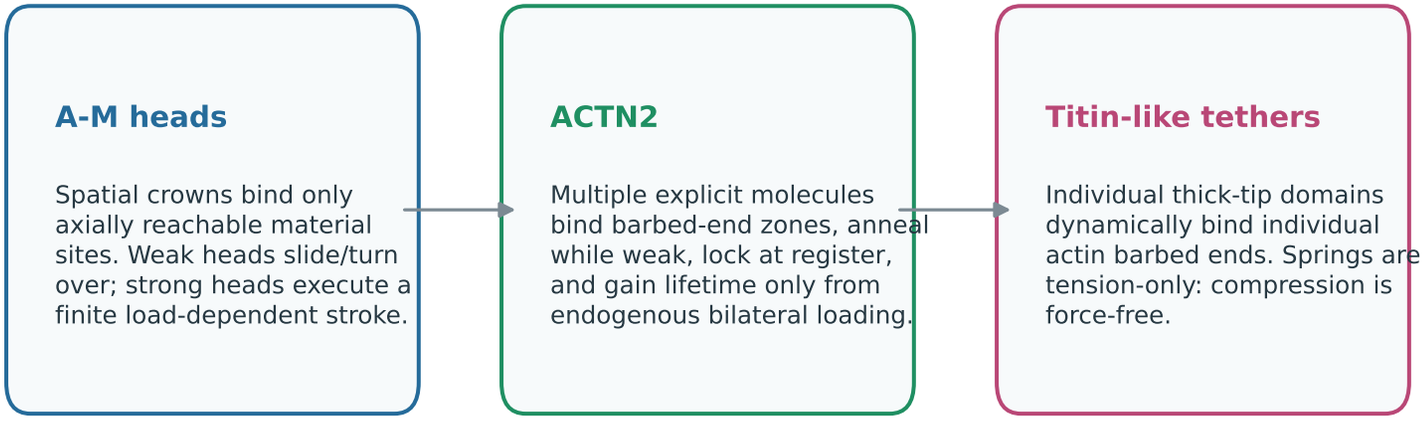
Current coarse-grained sarcomere model

Mechanistic summary and changes since the roughly 29 July working model · 5 August 2026

Goal: test whether myosin motors, turnover-capable ACTN2 and individual titin-like tethers can assemble and maintain preferred A-A register while preserving finite, dynamic A-M engagement—without prescribing a Z-disc node or an external load.



- **Dynamics**

  Anisotropic overdamped Brownian translation and rotation of finite rods; all molecular forces and torques are reciprocal; filament sterics prevent interpenetration.

- **No imposed stall**

  The force-velocity curve is sampled from actual ACTN2 connector load and actual sliding. Balanced motors slow because their working strokes load the crosslinked network.

- **No pre-existing Z-disc**

  There is no six-actin centroid, Z-node, prescribed ring radius or centroid-to-myosin spring. Any local bundle/register must emerge from molecular interactions.

Evidence included in this packet

Production-scale 3-D views use only the successful frozen model-19 runs: 1,280 actins, 640 myosins, 100 ms, 35 heads per half, alignment stiffness 120, seeds 2080201 and 2080202. Later collapsed-retention production runs are excluded. These runs support A-A maturation and nematic alignment; they are not presented as proof that A-M placement is solved.

Brownian rods and explicit A-M heads

Actin and myosin are finite spherocylindrical rods. Their mobility is anisotropic and derived from rod geometry, solvent viscosity and temperature.

$$d\mathbf{r} = \mathbf{M}_t\mathbf{F}\,dt + \sqrt{2k_BT\mathbf{M}_t}\,d\mathbf{W}$$

$$d\mathbf{u} = \left[\mathbf{M}_r\boldsymbol{\tau}\,dt + \sqrt{2k_BT\mathbf{M}_r}\,d\mathbf{W}_r\right] \times \mathbf{u}$$

Each myosin half owns spatial head anchors. A detached head can bind actin i only when its anchor projects inside the finite actin segment, lies on the productive myosin half, and is within surface capture reach:

$$|s_i^*| \le L_A/2, \qquad h_{head} = h_{prod}, \qquad d_{surf} \le d_c$$

$$k_{on}^{hi} = k_{on}^0\, a_i(t) \exp\left[-\frac{(d_{surf}-d_0)^2}{2\sigma_d^2}\right]$$

The bound actin coordinate is material and remains fixed until detachment. This prevents heads outside an actin from grabbing its endpoint and makes relative sliding generate strain.

$$D \longrightarrow W \longrightarrow S, \qquad W \longrightarrow D, \qquad S \longrightarrow D$$

$$F_\parallel = \min[k_m(q-z)_+, F_s], \qquad \dot{q} = v_0\left(1 - \frac{F_{resist}}{F_s}\right)_+$$

$$U_\perp = \tfrac{1}{2}k_\perp\,[\rho - \rho_0 - \delta_{slack}]_+^2$$

$$k_{off} = k_{chem}(F)\exp\left[\beta(\theta_\parallel U_\parallel^{np} + \theta_\perp U_\perp)\right]$$

The radial spring is a passive, slack tether on each bound head, not a permanent A-M bond. It appears only while that head is weak or strong and disappears on detachment. The final geometric rupture is an emergency contour guard, not the normal turnover mechanism.

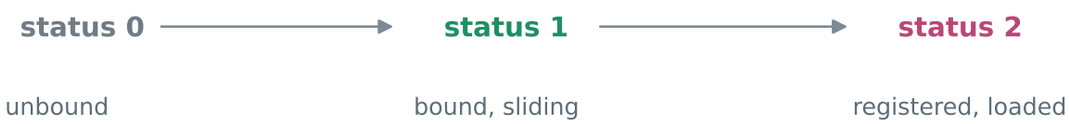
Explicit ACTN2: annealing, register and endogenous load

An ACTN2 molecule stores two material coordinates on antiparallel actins. Its connector acts in full 3-D, so it constrains both axial register and transverse separation.

$$\mathbf{c}_\ell = (\mathbf{r}_j + s_j\mathbf{u}_j) - (\mathbf{r}_i + s_i\mathbf{u}_i), \qquad r_\ell = |\mathbf{c}_\ell|$$

$$U_{AA,\ell} = \tfrac{1}{2}k_{r,\ell}(r_\ell - r_0)^2 + U_{register} + U_{angle}$$

Status 1 is a weak encounter: sites can anneal within the 200 nm barbed-end association zones. Status 2 is a registered, material-site lock near the preferred 0.10 micrometre A-A overlap. Association/unbinding and status are distinct variables.



The load used for maturation is not an applied force and is not inferred from overlap. Productive motor force on each actin is allocated across the ACTN2 arms capable of transmitting it:

$$A_{i\ell} = M_i \frac{R_{i\ell}}{\sum_m R_{im}}, \qquad \widetilde{L}_{i\ell} = \min(R_{i\ell}, A_{i\ell})$$

$$L_\ell = \min(\widetilde{L}_{i\ell}, \widetilde{L}_{j\ell})$$

$$\text{status 2} = \text{registered} \times \text{productive opposed motor support}$$

$$\tau_\ell = \tau_0 + \mathcal{I}_{s=2}(\tau_{dir} + \lambda L_\ell), \qquad P_{off} = \min(dt/\tau_\ell, 1)$$

A separate contour test removes a connector if its actual 3-D length exceeds the rupture limit. This split lets us distinguish stochastic chemistry from geometric failure in diagnostics.

Explicit titin-like molecules—without a Z-node

Each thick-filament tip carries explicit binding domains. A domain binds one individual actin barbed end, turns over, and can rebind. It never refers to a six-actin centroid.

$$x_T = [d_T - \ell_T]_+$$

$$U_T = \tfrac{1}{2} k_T x_T^2, \qquad F_T = k_T x_T$$

Compression therefore produces no force. A tether becomes mechanically active only after its end-to-end distance exceeds the slack length.

$$k_{on,T} = k^0_{on,T}\, \mathcal{I}(d_T \le d_c)\, \mathcal{I}(\text{live ACTN2 support})$$

$$k_{off,T} = k^0_{off,T} \exp[\beta(1-\theta_T)U_T]$$

The support gate is a formation/rebinding rule, not a centroid constraint: titin is licensed by a locally crosslinked barbed end, then applies equal and opposite force to that actin and the individual myosin-tip domain. Unsupported serial anchors can release quickly.

- **What titin fixes**

  It removes the common-translation zero mode of an A-A pair by coupling individual barbed ends to the thick-filament lattice while remaining slack under compression.

- **What titin does not assume**

  No Z-disc position, ring radius, six-actin membership, A-A centroid, or preferred azimuth is supplied to the force law.

- **Remaining coarse graining**

  The binding domain and spring stand in for a titin-like axial tether; actin polymerization and the molecular architecture of a mature Z-disc are not modeled.

What changed over the past week

Mechanism	~29 July working model	Current model
A-M capture	Broad pair/end-point reach	Material crowns; anchor projection must lie inside finite actin an
Motor mechanics	Overlap/status proxy	Weak/strong heads, fixed material sites, finite stroke, radial teth
A-A model	One pair-level spring	Multiple explicit ACTN2 molecules; weak annealing and registere
Status 2	cb=2 plus substantial overlap / mot	Registry plus productive opposed motor support; load comes fro
Titin	Occupancy/centroid-triggered spring	Individual dynamic tip-to-barbed-end molecules; tension-only; n
Motion	Simpler translational noise	Anisotropic physical rod mobility, rotational diffusion, torques an
Balanced F-V	Applied diagnostic load	Endogenous ACTN2 load paired with measured sliding velocity
Assembly prior	Implicit Z-node/centroid stabilizers i	Removed: local geometry must emerge from molecular interacti

Successful older scale-up used for the 3-D explorer

seed 2080201 342 active ACTN2 links · 185 status-2 actins · S_A = 0.838

seed 2080202 322 active ACTN2 links · 186 status-2 actins · S_A = 0.893

Interpretation: the run demonstrates robust nematic alignment and maturation near the preferred A-A register. Finite A-M placement remains a model question and is deliberately not claimed here. No result from the later collapsed-ACTN2-retention production trajectories appears in this document.